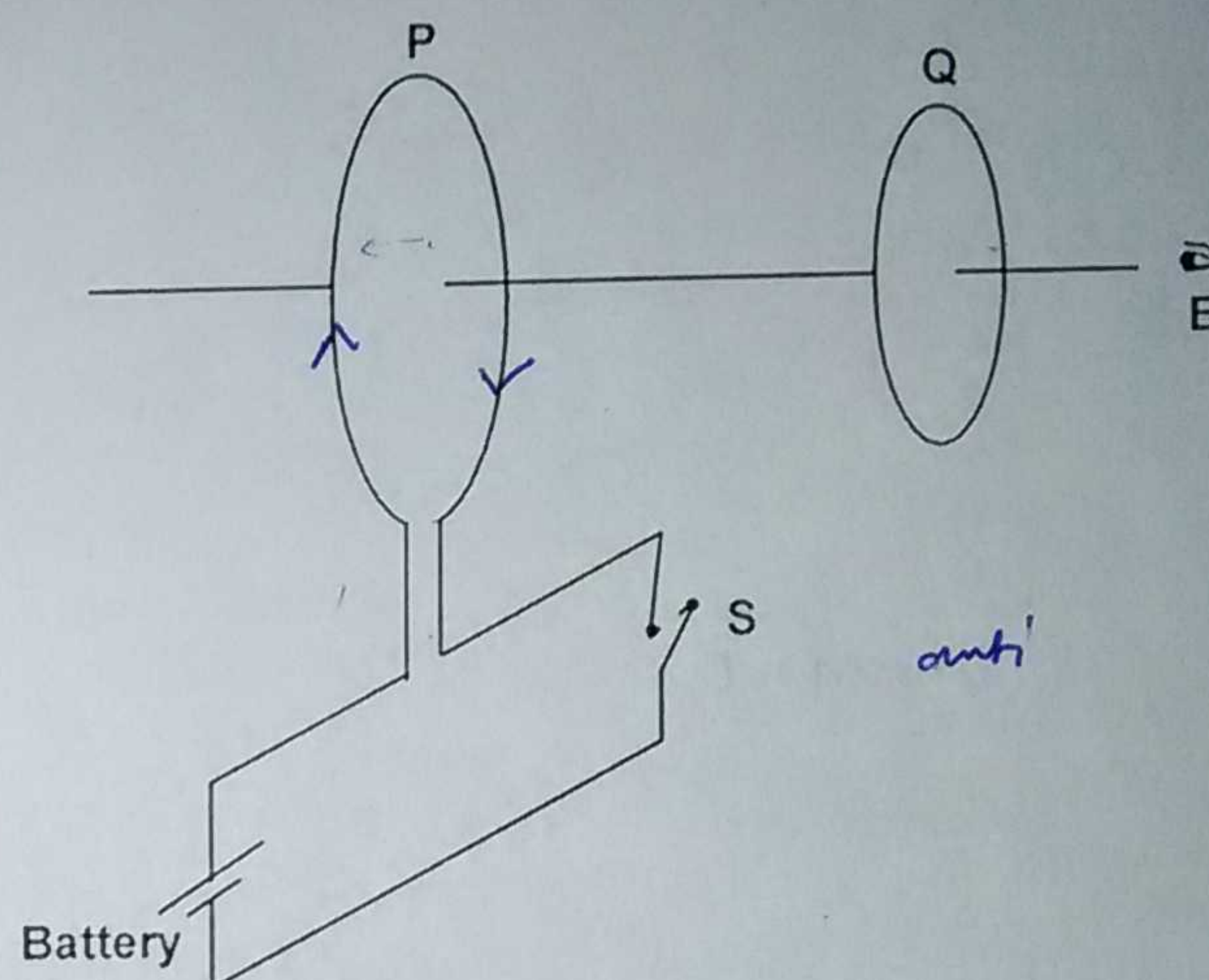


Electromagnetic Induction - I

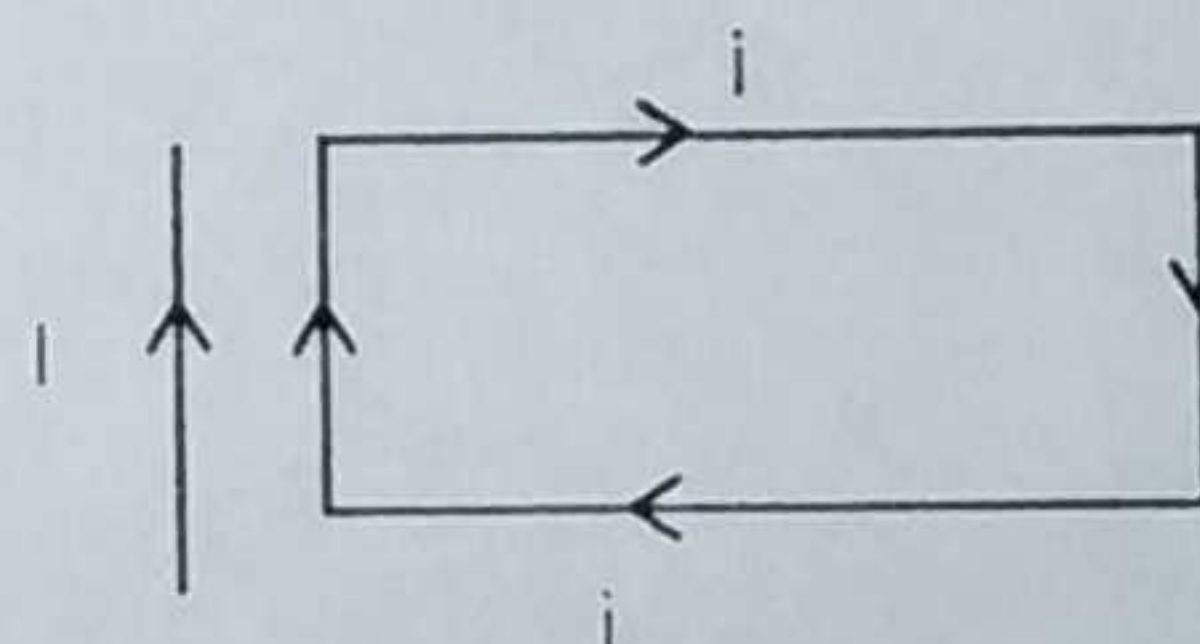
1. A rod of the length L rotates with a uniform angular velocity ω about an axis passing through mid point. The magnetic field B exists parallel to the axis of rotation. The emf between the centre of the rod and an end is
 (A) zero ☒ (B) $\frac{1}{8} \omega B L^2$ (C) $\frac{1}{2} \omega B L^2$ (D) $B \omega L^2$
2. A short-circuited coil is placed in a time-varying magnetic field. Electrical power is dissipated due to the current induced in the coil. If the number of turns was to be quadrupled and the wire radius halved, the electrical power dissipated would be
 (A) halved (B) the same. ☒
 (C) doubled (D) quadrupled

3. As shown in the figure, P and Q are two coaxial conducting loops separated by some distance. When the switch S is closed, a clockwise current I_P flows in P (as seen by E) and an induced current I_{Q1} flows in Q. The switch remains closed for a long time. When S is opened, a current I_{Q2} flows in Q. Then the direction I_{Q1} and I_{Q2} (as seen by E) are



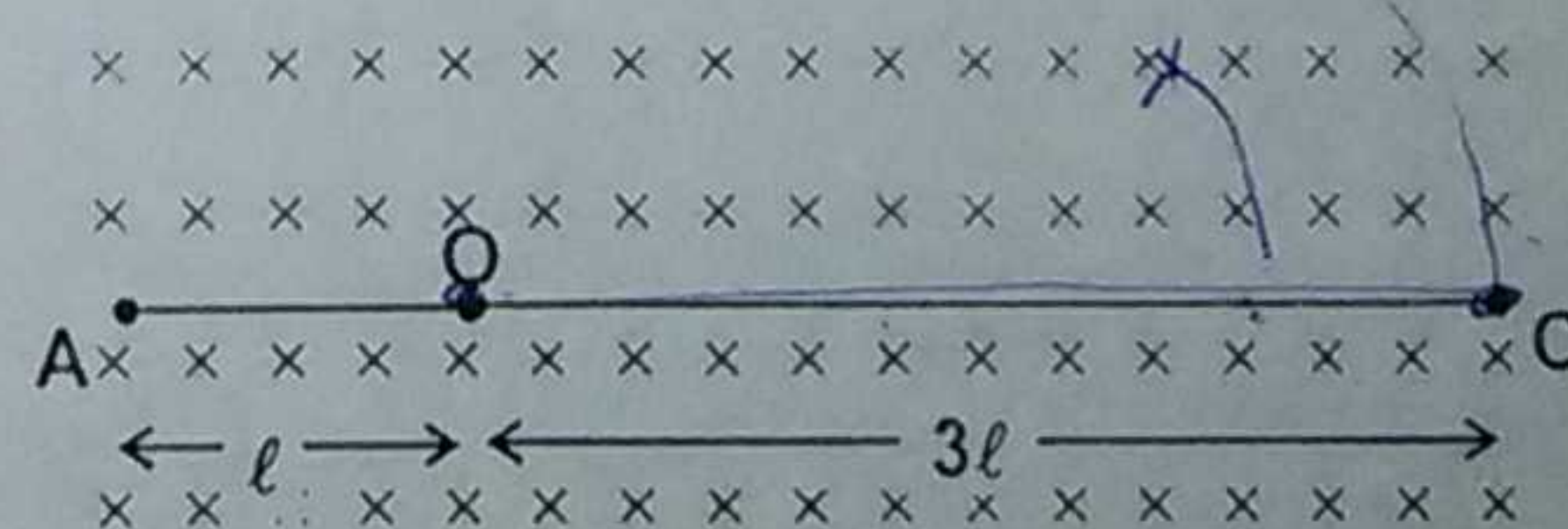
- (A) respectively clockwise and anti-clockwise.
 (B) both clockwise.
 (C) both anti-clockwise.
☒ (D) respectively anti-clockwise and clockwise.

4. A rectangular loop carrying current I is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and is in the plane of the loop. If a steady current I is established in the wire as shown in figure the loop will



- (A) rotate about an axis parallel to the wire (B) move away from the wire
☒ (C) move towards the wire (D) remain stationary

5. A conduction rod AC of length ' 4ℓ ' is rotated about a point O in a uniform magnetic field $\vec{B}$ directed in the plane of paper. Take $AO = \ell$, $OC = 3\ell$ then

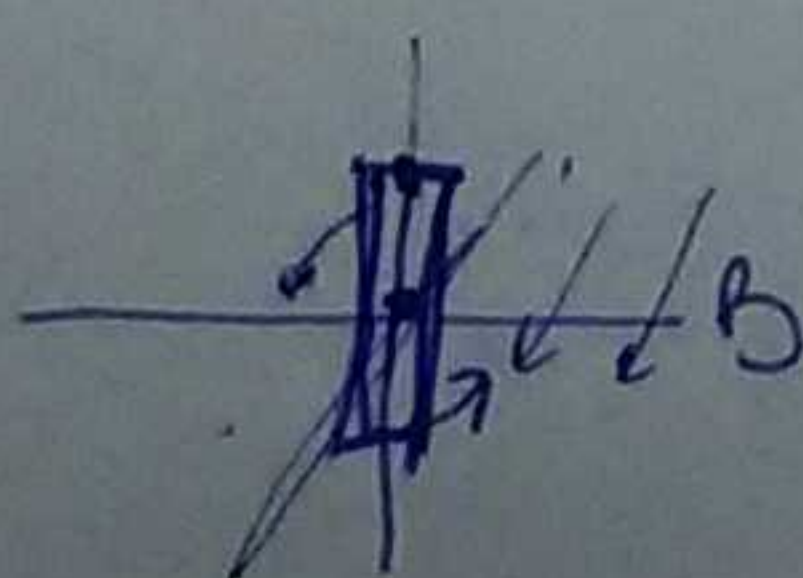


(A) $V_C - V_O = \frac{B \omega \ell^2}{2}$

☒ (B) $V_O - V_C = \frac{9}{2} B \omega \ell^2$

☒ (C) $V_C - V_O = \frac{9}{2} B \omega \ell^2$

(D) none of these.



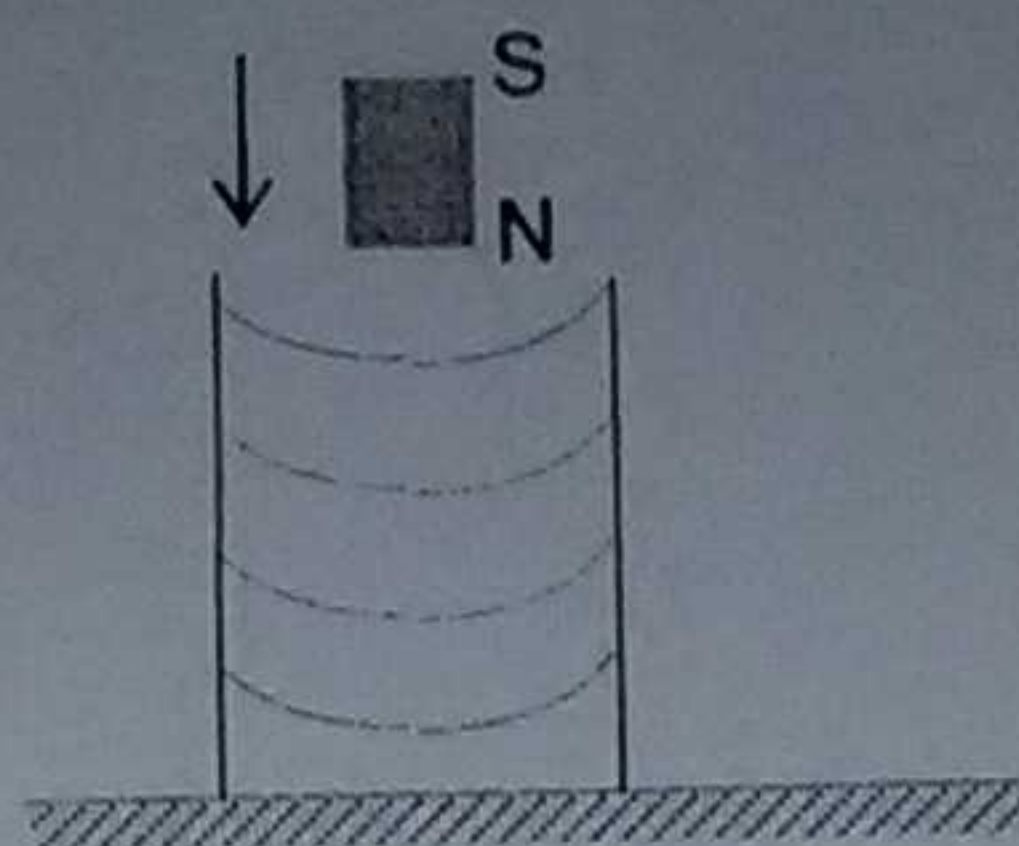
$$\omega \times B = E$$

$$\left[\frac{\omega \times^2 B}{2} \right]_0^{4\ell/2}$$

$$\frac{\omega \times^2 B}{2}$$

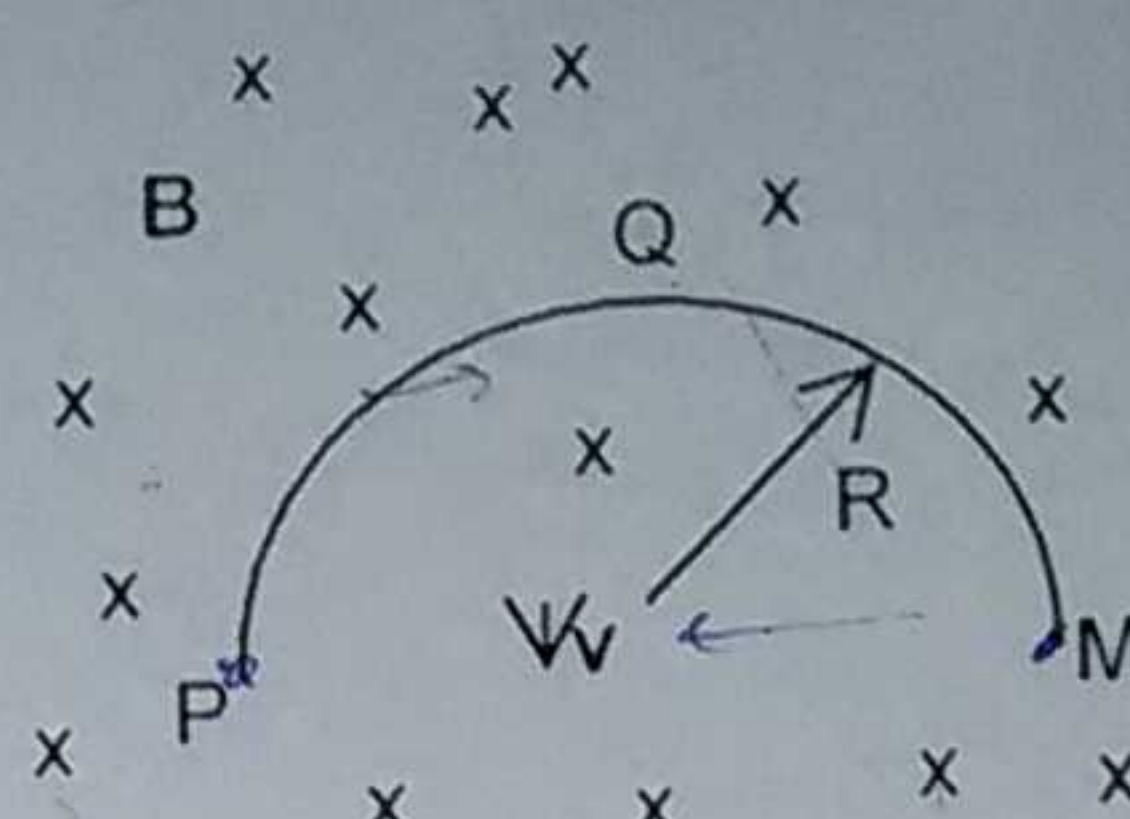
$$\frac{\omega \times (3\ell)^2 B}{2}$$

6. A magnet placed vertically is falling freely over a cylindrical coil placed over a horizontal surface as shown. The acceleration of the magnet is



- (A) equal to g (B) greater than g (C) less than g (D) directed upwards.

7. A conducting thin semicircular ring of radius R falls in a horizontal magnetic induction B such that $\vec{v} \cdot \vec{B} = 0$. At the position PQM, the speed of ring is v . The potential difference developed across the ring is



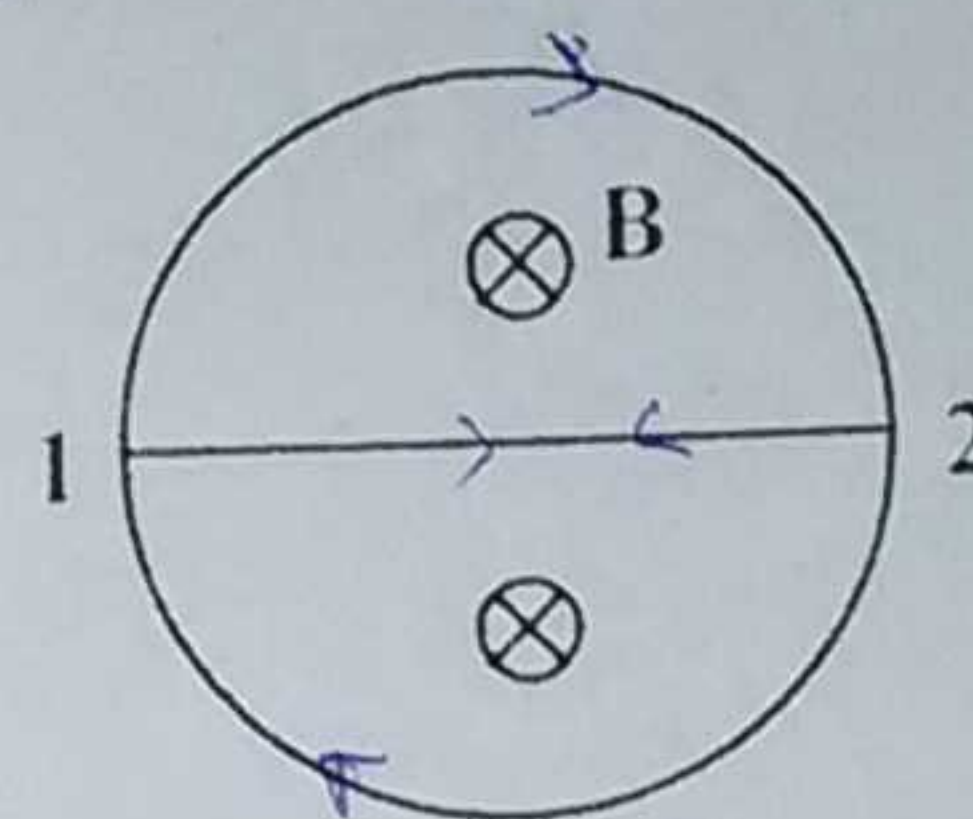
$$\frac{qVB}{r} = E$$

1/

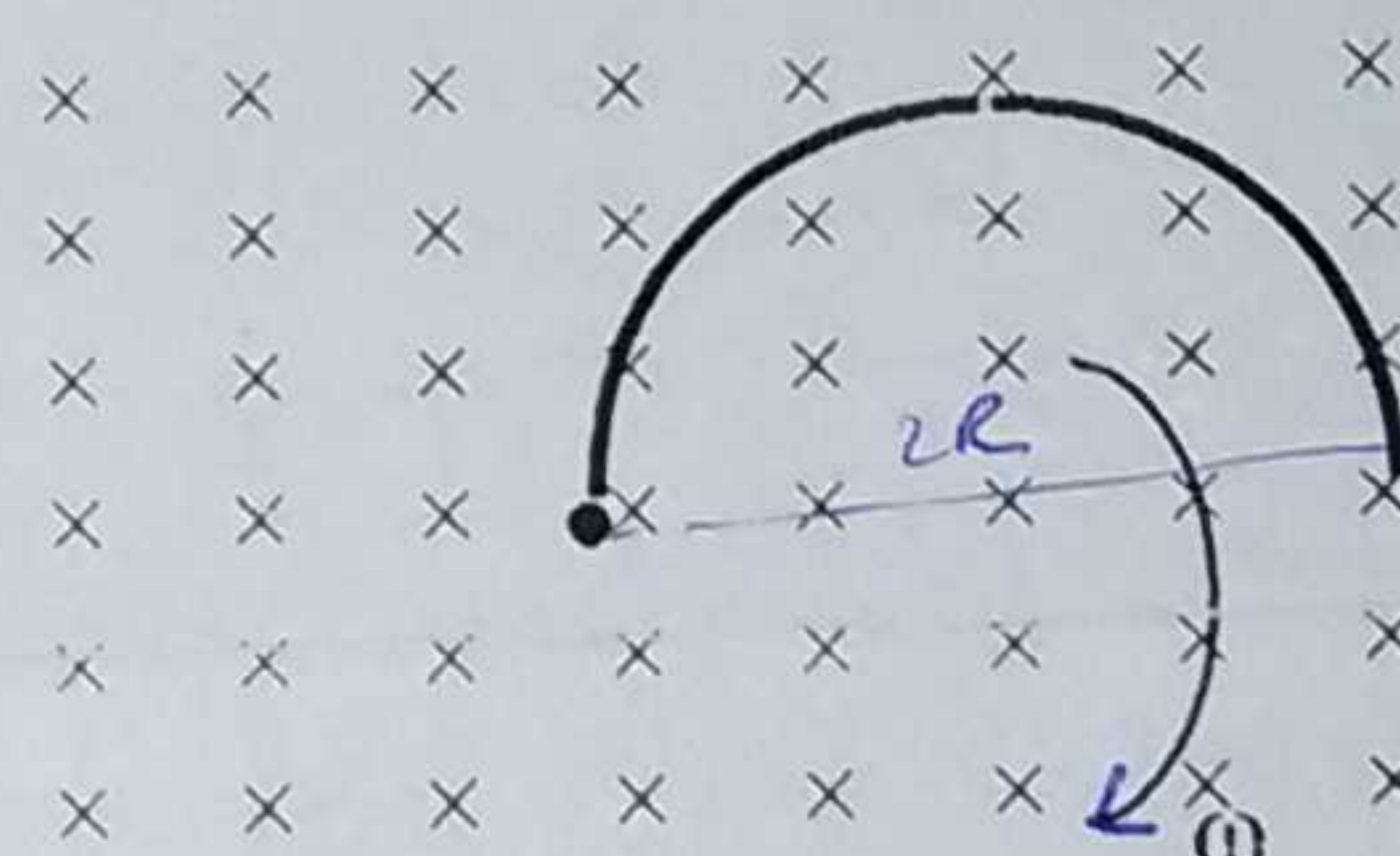
- (A) πRBv , M at higher potential (B) $2RBv$, M at higher potential
(C) zero (D) $2RBv$, P at higher potential

8. The diametric opposite points (1 & 2) of a conducting ring are joined together. If the magnetic field B starts diminishing with time, indicate the direction of current in the wire 1-2.

ans. 0



9. A semicircular conducting rod of radius r is rotated about one of its free end in a uniform magnetic field B at constant angular velocity ω . Emf induced across its ends is equal to

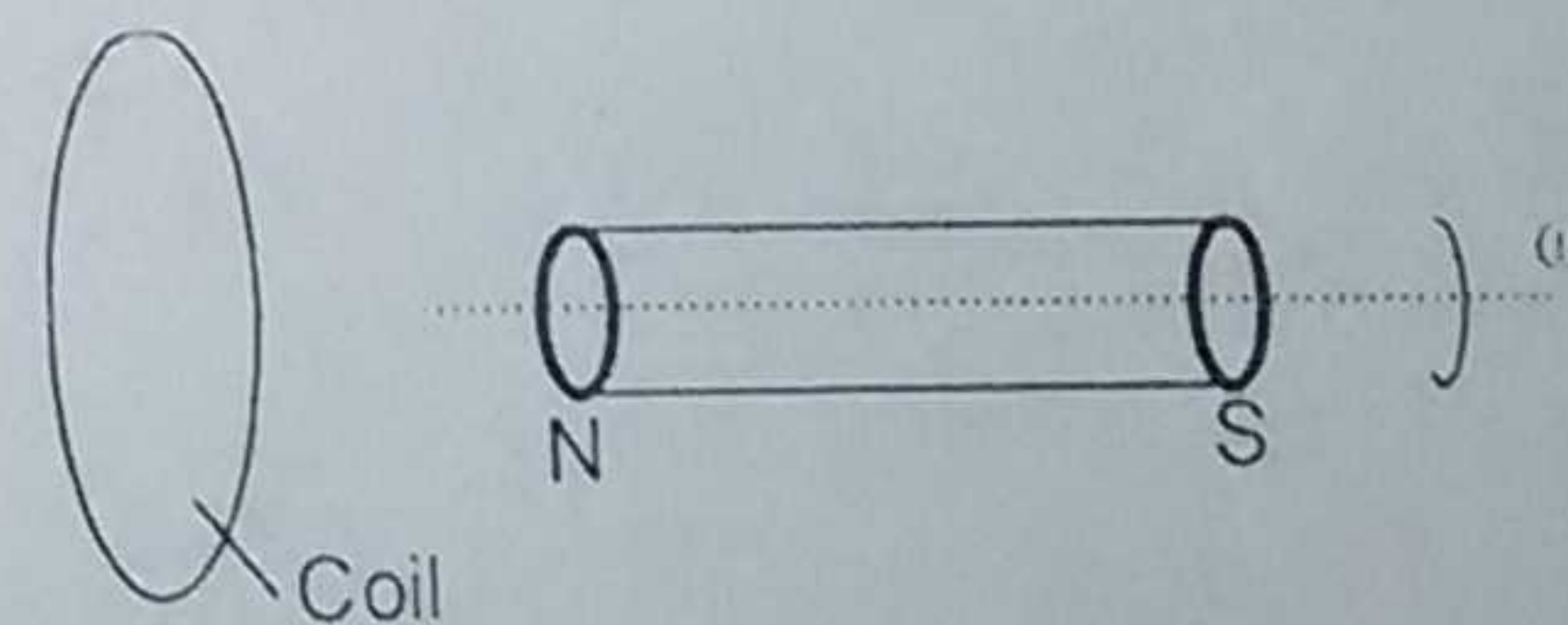


- (A) zero (B) $2B\omega R^2$
(C) $(1/2)B\omega R^2$ (D) $B\omega R^2$

10. A cylindrical magnet is kept along the axis of a circular coil. If magnet is rotated about the axis then

- (A) current is induced in the coil. (B) only e.m.f. is induced in the coil
(C) both emf and current are induced in the coil. (D) None of the above.

11. A cylindrical magnet is placed near a circular coil. The magnet is rotated about its own axis as shown in figure. The induced current in the coil is :



- (A) zero (B) in anticlockwise direction
(C) in clockwise direction (D) directly proportional to the speed of rotation ω
12. Two identical coaxial circular loops carry a current i , each circulating in the same direction. If the loops approach each other you will observe that

- (A) the current in each increases
(B) the current in each decreases
(C) the current in each remains the same
(D) the current in one increases whereas the current in the other decreases